
AR 505: Principles of Robot Autonomy

Course Code	AR 505
Course Name	Principles of Robot Autonomy
L-T-P-C	3-0-0-3
Prerequisites	Consent of faculty advisor
Intended for	UG, PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction:** Brief overview along with motivation and potential applications. (1 hours)
- **Robotic Perception:** Robotic sensors, robotic sensor calibration and its importance, robot vision vs computer vision, robot localization, artificial neural networks for robot perception. (16 hours)
- **Robot Motion Planning:** Overview, Configuration space, Free space, Target space, obstacle space, traditional and machine learning based planning algorithms. (14 hours)
- **Robot control:** P, PI, PD and PID controller, visual servoing, and multi-robot control. (11 hours)

Text Books

1. Choset, Howie, Kevin M. Lynch, Seth Hutchinson, George A. Kantor, and Wolfram Burgard, **Principles of robot motion: theory, algorithms, and implementations**, MIT press, 2005.
2. Apolloni, Bruno, et al., (eds.), **Machine learning and robot perception**, Vol. 7. Springer Science & Business Media, 2005.

References

1. Cuesta, Federico, and Anibal Ollero, **Intelligent mobile robot navigatio**., Vol. 16. Springer Science & Business Media, 2005.
2. Steve LaValle, **Planning Algorithms**, Cambridge Univ. Press, 2006.
3. Mouha, Radouan Ait, Deep Learning for Robotics, **Journal of Data Analysis and Information Processing**, 9.02:63, 2021.